Section A – Answer any 2 out of 3 questions in section A

Question 1

a) There are two identical generators, each of 100 kW capacity and availability of 95%. Assume the reliability criterion requires that the probability to meet peak demand is above 92%, and the coincidence factor of an infinite number of households is 0.444. The peak demand of each individual household is 5 kW.

(i) how many households can be supplied by these two generators?

[4]

Pn =100kW.

[2 marks]

$$\left(0.444 + \frac{1-0.444}{\sqrt{n}}\right) n \times 5 = 100$$
$$\therefore n = 37.39$$
37 households

[2 marks]

(ii) how many additional households can be supplied if a 100 kW wind farm with 50% availability is connected to the system?

[4]

Pn =200kW.

[2 marks]

$$\left(0.444 + \frac{1-0.444}{\sqrt{n}}\right) n \times 5 = 200$$
$$\therefore n = 78.96$$
78 households
$$\Delta n = 78 - 37 = 41$$

[2 marks]

b) Battery storage systems can provide frequency response much faster than thermal generators, therefore providing a service called Enhanced Frequency Response (EFR) in the UK. EFR reduces the need for the regular frequency response (R), following this rule :

$$\left(\frac{H}{f_0} - \mathrm{EFR} \cdot \frac{\mathrm{T_{EFR}}}{4 \cdot \Delta f_{\max}}\right) \cdot R \geq \frac{(P_{\mathrm{L}} - \mathrm{EFR})^2 \cdot \mathrm{T_d}}{4 \cdot \Delta f_{\max}}$$

Nomenclature

H: system level of inertia (Units: MW · s).
R: amount of frequency response scheduled (Units: MW).
EFR: amount of EFR scheduled (Units: MW).
P_{L}: value of the generation outage we want to secure against (Units: MW).
f_0: nominal electric frequency of the power grid (Units: Hz).
$\Delta f_{\max}$: maximum admissible frequency deviation to avoid load shedding (Units: Hz).
$\mathrm{T_d}$: Delivery-time of frequency response (Units: s).
$\mathrm{T_{EFR}}$: Delivery-time of EFR (Units: s).

Model Answers

Assumptions:
1. The system level of inertia of 150 GWs, i.e. H = 150 GW-s,
2. The cost of frequency response: \$5/MW,
3. The largest generation unit in the UK: 1.8 GW,
4. Other numerical values: f_0 = 50 Hz, Δfmax = 0.8 Hz, T_d = 10s, T_{EFR} = 0.5 s.

If there is a bid of 200MW of EFR for £20/MW, should the system operator buy it to achieve the minimum cost?

1) Without EFR

R=3375MW, EFR =0,

[2 marks]

2) With EFR
R=2694MW, EFR=200MW,

[2 marks]

cost 1= 16875
cost 2= 17473

Since cost1<cost2, the operator should not buy it.

[2 marks]

[6]

c) In a balanced three-phase star-connected system, as shown in Figure 1, write down the expressions for current through the neutral connection I_n as functions of voltage magnitudes and phase angles.

[6]

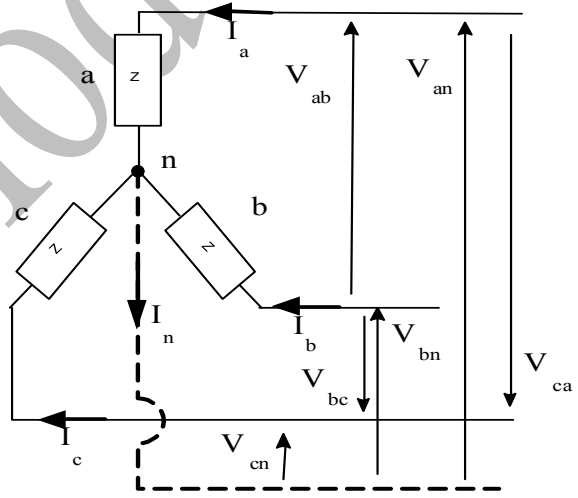


Figure 1 Balanced three-phase star-connected system

$$v_a(t) = \sqrt{2}V sin\omega t\ ,\ i_a(t) = \frac{\sqrt{2}V}{z} sin(\omega t - \phi)$$

Model Answers

$v_b(t) = \sqrt{2}V sin(\omega t - 120^o)$, $i_b(t) = \frac{\sqrt{2}V}{Z} sin(\omega t - \phi - 120^o)$

$v_b(t) = \sqrt{2}V sin(\omega t - 240^o)$, $i_c(t) = \frac{\sqrt{2}V}{Z} sin(\omega t - \phi - 240^o)$

[3 marks]

Phasor expression:

$I_a = \frac{V}{Z}\angle -\phi$, $I_b = \frac{V}{Z}\angle -\phi - 120$, $I_c = \frac{V}{Z}\angle -\phi - 240$

$I_n = I_a + I_b + I_c = 0$

[3 marks]

Question 2

a) What are the principles in selecting the base value for per-unit analysis in power systems?

[3]

Choose ONE base power for the entire system
- Arbitrary choice
- Commensurate with the system
- e.g. 100 MW, not 100 mW

Choose ONE base voltage for each voltage level
- Choose the nominal voltage

Other base values chosen to get the same relations between per unit quantities as between actual quantities

[1 mark each]

b) Consider a high voltage transmission circuit in Figure 2..1 Assume that voltages (magnitudes and phase angles) are known at both sending and receiving ends.

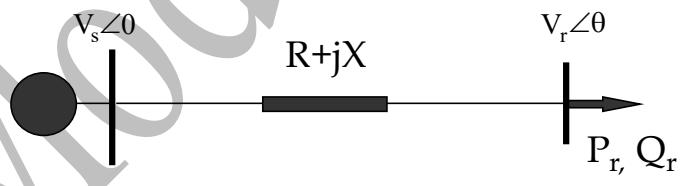


Figure 2.1 Transmission circuit with voltages specified at the both sending and receiving ends

(i) Write down the expressions for sending end active and reactive powers as functions of voltage magnitudes and phase angles.

[3]

$$S = V_s i_s^* = V_s\left(\frac{V_s - V_r\angle\theta}{R + jX}\right)^* = V_s\left(\frac{V_s - V_r\cos\theta - jV_r\sin\theta}{R + jX}\right)^*$$
$$= \frac{V_s^2 R - V_s V_r(R\cos\theta + X\sin\theta)}{R^2 + X^2} + j\frac{V_s^2 X + V_s V_r(R\sin\theta - X\cos\theta)}{R^2 + X^2}$$
$$= P_s + jQ_s$$

Model Answers

(ii) Based on these expressions and assuming R>>X, show that active power flow requires a difference in voltage magnitude and that reactive power flow requires a difference in phase angle.

[3]

$$P_s = \frac{V_s^2 - V_s V_r \cos\theta}{R}, \qquad Q_s = \frac{V_s V_r \sin\theta}{R}$$

c) There is a three-phase system as shown in Figure 2.2. A generator with a capacity of 60MVA is connected to a 15kV bus, which is fed into a 11kV substation. The load receives active power 36MW and reactive power 27MVA. Calculate the voltage at the receiving end (V_r) and the complex power (P_s+jQ_s) at the sending end given $S_B = 50\text{MVA}$.

[11]

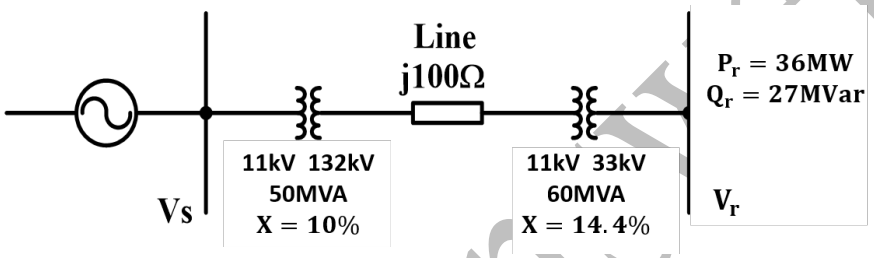


Figure 2.2 Three-phase system is supplied by a generator

Choose: $V_{B11} = 11kV,\ V_{B132} = 132kV, V_{B33} = 33kV$

[1 mark]

$P_r = 0.72p.u. \quad Q_r = 0.54p.u.$

[1 mark]

$$X = X_1 + X_2 + X_3 = j0.1 + \frac{j100}{V_{B132}^2/S_B} + j14.4\frac{50}{60} = j0.507p.u.$$

[1 mark]

$$V_s = V_r + ji_rX = V_r + \frac{j(P_r - jQ_r)}{V_r^*}X$$

$$(V_s - V_r)V_r^* = j(P_r - jQ_r)X$$

$$V_s = \frac{15}{11}\angle 0° = 1.3636\angle 0° \quad V_r = |V_r|\angle\theta$$

$$V_s V_r \cos\theta - |V_r|^2 = Q_r X; \quad V_s V_r \sin\theta = -P_r X$$

[2 marks]

Eliminate the angle: $|V_r|^4 + (2Q_rX - V_s^2)|V_r|^2 + (Q_r^2 + P_r^2)X^2 = 0$

$$\therefore |V_{r1}| = 1.0616p.u. \qquad \therefore |V_{r2}| = 0.4298p.u.$$

[2 marks]

$$P_r = \frac{V_s V_r}{X}\sin(-\theta)$$

Model Answers

$$\theta_1 = -sin^{-1}\frac{P_r X}{V_s|V_{r1}|} = -sin^{-1}\frac{0.72\times j0.507}{1.3636\times 1.0616} = -14.61°$$

$$\theta_2 = -sin^{-1}\frac{P_r X}{V_s|V_{r2}|} = -sin^{-1}\frac{0.72\times j0.507}{1.3636\times 0.4298} = -38.52°$$

[1 mark]

$$i_1 = \frac{S_r^*}{V_{r1}^*} = \frac{0.72-j0.54}{1.0616\angle 14.61°} = 0.8478\angle -51.48°$$

$$i_2 = \frac{S_r^*}{V_{r2}^*} = \frac{0.72-j0.54}{0.4298\angle 38.52°} = 2.0940\angle -75.39°$$

[1 mark]

$S_{s1} = V_s i_1^* = 1.3636 \times 0.8478\angle 51.48° = 0.72 + j0.9045 p.u. = 36MW + 45.2245MVar$

$|S_{s1}| = 57.8036MVA < 60MVA$

$S_{s2} = V_s i_2^* = 1.3636 \times 2.0940\angle 75.39° = 0.72 + j2.7631 p.u. = 36MW + 138.1525MVar$

$|S_{s2}| = 142.7656MVA \gg 60MVA$, *infeasible.*

$\therefore V_r = 1.0616\angle -14.61° p.u. = 35.0328\angle -14.61° kV$

$P_s + jQ_s = 36MW + j45.2245MVar$

[2 marks]

Model Answers

Question 3

A four-bus power network is presented in Figure 3 below. Data relevant for the load flow analysis on this system are given in per-unit values.

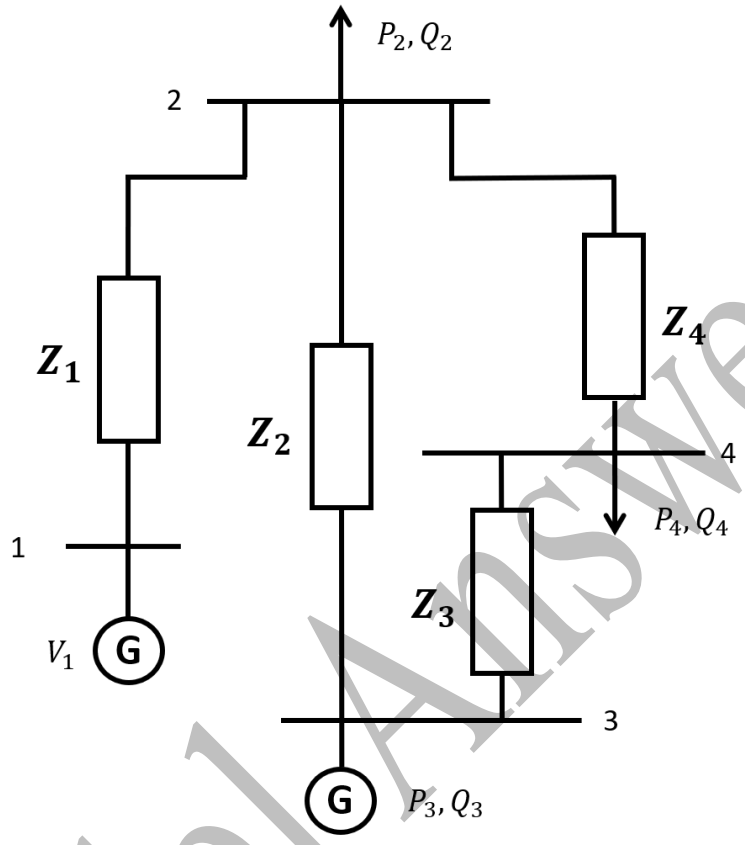


Figure 3. A four-bus power network

a) Given Z_1, Z_2, Z_3 and Z_4, form the bus admittance matrix [6]

$$i_1 = \frac{1}{Z_1}(V_1 - V_2) = \frac{1}{Z_1}V_1 - \frac{1}{Z_1}V_2$$

$$i_2 = \frac{1}{Z_1}(V_2 - V_1) + \frac{1}{Z_2}(V_2 - V_3) + \frac{1}{Z_3}(V_2 - V_4)$$

$$= -\frac{1}{Z_1}V_1 + \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}\right)V_2 - \frac{1}{Z_2}V_3 - \frac{1}{Z_3}V_4$$

$$i_3 = \frac{1}{Z_2}(V_3 - V_2) + \frac{1}{Z_3}(V_3 - V_4) = -\frac{1}{Z_2}V_2 + \left(\frac{1}{Z_2} + \frac{1}{Z_4}\right)V_3 - \frac{1}{Z_4}V_4$$

$$i_4 = \frac{1}{Z_3}(V_4 - V_2) + \frac{1}{Z_4}(V_4 - V_3) = -\frac{1}{Z_3}V_2 - \frac{1}{Z_4}V_3 + \left(\frac{1}{Z_3} + \frac{1}{Z_4}\right)V_4$$

[3 marks]

Model Answers

$$\therefore Y = \begin{pmatrix} \frac{1}{Z_1} & -\frac{1}{Z_1} & 0 & 0 \\ -\frac{1}{Z_1} & \frac{1}{Z_1}+\frac{1}{Z_2}+\frac{1}{Z_3} & -\frac{1}{Z_2} & -\frac{1}{Z_3} \\ 0 & -\frac{1}{Z_2} & \frac{1}{Z_2}+\frac{1}{Z_4} & -\frac{1}{Z_4} \\ 0 & -\frac{1}{Z_3} & -\frac{1}{Z_4} & \frac{1}{Z_3}+\frac{1}{Z_4} \end{pmatrix}$$

[3 marks]

b) Assuming the per unit values $|V_1| = 1$, $|V_3| = 1.1$, $P_3 = 1.1$, $P_2 = 0.95$, $Q_2 = 1$, $P_4 = 1.05$, $Q_4 = 1$, $Z_1 = 0.25j$, $Z_2 = 0.25j$, $Z_3 = 0.2j$ and $Z_4 = 0.2j$, perform two iterations of the Gauss-Seidel load flow method.

[10]

$V_1 = 1\angle 0°$ $\quad V_2^0 = 1\angle 0°$ $\quad V_3^0 = 1.1\angle 0°$ $\quad V_4^0 = 1\angle 0°$

[1 mark]

$$Y = \begin{pmatrix} -j4 & j4 & 0 & 0 \\ j4 & -j13 & j5 & j4 \\ 0 & j5 & -j10 & j5 \\ 0 & j4 & j5 & -j9 \end{pmatrix}$$

[1 mark]

Iteration 1:

$$i_2 = \frac{S_2^*}{V_2^*} = Y_{21}V_1 + Y_{22}V_2 + Y_{23}V_3 + Y_{24}V_4$$

$$V_2^1 = \frac{1}{Y_{22}}\left(\frac{S_2^*}{V_2^{0*}} - Y_{21}V_1 - Y_{23}V_3^0 - Y_{24}V_4^0\right) = \frac{1}{-j13}\left(\frac{-0.95+j}{1} - j4 - j5 \times 1.1 \ - j4\right)$$
$$= 0.9643\angle -4.3461°$$

[1 mark]

$$S_3^* = V_3^{0*} i_3 = V_3^{0*}(Y_{31}V_1 + Y_{32}V_2^1 + Y_{33}V_3^0 + Y_{34}V_4^0)$$
$$= 1.1 \times (0 + j5 \times (0.9643\angle -4.3461°) - j10 \times 1.1 \ + j5)$$
$$= 0.4019 - j1.3115$$

$$S_3^{1*} = 1.1 - j1.3115$$

$$V_3^1 = \frac{1}{Y_{33}}\left(\frac{S_3^{1*}}{V_3^{0*}} - Y_{31}V_1 - Y_{32}V_2^1 - Y_{34}V_4^0\right)$$
$$= \frac{1}{-j10}\left(\frac{1.1 - j1.3115}{1.1} - 0 - j5(0.9643\angle -4.3461°) - j5\right)$$
$$= 1.1018\angle 3.3019°$$

$$V_3^1 = 1.1\angle 3.3019°$$

[2 marks]

$$V_4^1 = \frac{1}{Y_{44}}\left(\frac{S_4^*}{V_4^{0*}} - Y_{41}V_1 - Y_{42}V_2^1 - Y_{43}V_3^1\right)$$
$$= \frac{1}{-j9}\left(\frac{-1.05+j}{1} - 0 - j4(0.9643\angle -4.3461°) - j5(1.1\angle 3.3019°)\right)$$
$$= 0.9331\angle -7.0139°$$

[1 mark]

Model Answers

Iteration 2:

$$V_2^2 = \frac{1}{Y_{22}}\left(\frac{S_2^*}{V_2^{1*}} - Y_{21}V_1 - Y_{23}V_3^1 - Y_{24}V_4^1\right)$$
$$= \frac{1}{-j13}\left(\frac{-0.95 + j}{0.9643\angle 4.3461°} - j4 - j5 \times 1.1\angle 3.3019° - j4(0.9331\angle -7.0139°)\right) = 0.9333\angle -4.9317°$$

[1 mark]

$$S_3^* = V_3^{1*} i_3 = V_3^{1*}(Y_{31}V_1 + Y_{32}V_2^2 + Y_{33}V_3^1 + Y_{34}V_4^1)$$
$$= 1.1\angle -3.3019° \times \left(0 + j5 \times (0.9333\angle -4.9317°) - j10 \times (1.1\angle 3.3019°) + j5(0.9331\angle -7.0139°)\right) = 1.6540 - j1.9696$$
$$S_3^{2*} = 1.1 - j1.9696$$
$$V_3^2 = \frac{1}{Y_{33}}\left(\frac{S_3^{2*}}{V_3^{1*}} - Y_{31}V_1 - Y_{32}V_2^2 - Y_{34}V_4^1\right)$$
$$= \frac{1}{-j10}\left(\frac{1.1 - j1.9696}{1.1\angle -3.3019°} - 0 - j5(0.9333\angle -4.9317°) - j5(0.9331\angle -7.0139°)\right) = 1.1011\angle 0.6786°$$
$$V_3^2 = 1.1\angle 0.6786°$$

[2 marks]

$$V_4^2 = \frac{1}{Y_{44}}\left(\frac{S_4^*}{V_4^{1*}} - Y_{41}V_1 - Y_{42}V_2^2 - Y_{43}V_3^2\right)$$
$$= \frac{1}{-j9}\left(\frac{-1.05 + j}{0.9331\angle 7.0139°} - 0 - j4(0.9333\angle -4.9317°) - j5(1.1\angle 0.6786°)\right) = 0.9364\angle -10.2734°$$

[1 mark]

c) List the possible short- and long-term options if the power flow across a transmission line exceeds its rated capacity.

- Short term
  - Re-dispatch generation
- Long term
  - Build more lines
  - Build lines with higher nominal voltage
  - Encourage location of new generation closer to the load centre

[1 mark each point]

[4]

Model Answers

Section B – Answer any 2 out of 3 questions in section B

Question 4

a) Justify or rectify the following statements with proper explanation:

(i) The bus impedance matrix used for fault analysis can be obtained by inverting the bus admittance matrix that is used for power flow analysis.

[2]

- The bus impedance matrix used for power flow analysis does not consider the reactance (sub-transient or otherwise) of the generators.
- For fault analysis, sub-transient reactance of the generators needs to be included in the bus admittance matrix before inverting it to get the bus impedance matrix although this is not necessarily the most efficient way for large systems, in particular.

[1 mark each]

Most students missed the main point that the bus admittance matrix used in power flow analysis does not include the generator sub-transient reactance which is required for fault analysis. Some students mistakenly referred to the issue of invertibility of the bus admittance matrix.

(ii) The flux linkage in a closed circuit with finite electromotive force and finite resistance cannot change instantaneously.

[2]

$$\sum Ri + \frac{d\psi}{dt} = \sum e$$

Integrating over a finite time interval Δt,

$$\sum R \int_0^{\Delta t} i\,dt + \Delta\psi = \sum \int_0^{\Delta t} e\,dt$$

For finite R and e as $\Delta t \to 0$, $\Delta\psi \to 0$

[1 mark each]

(iii) It is completely unreasonable to assume the pre-fault line currents to be zero in order to simplify fault calculations.

[2]

It is reasonable to assume the pre-fault line currents to be zero because

- Line currents during fault are typically much larger than the pre-fault line currents
- Line currents during fault are predominantly reactive while the pre-fault line currents are predominantly resistive in nature which leads to a relatively small error if the latter are neglected.

[1 mark each]

Some students missed the 2nd point

Model Answers

(iv) The short circuit level of a power system reduces when conventional synchronous generation is replaced by wind and solar photovoltaic generation.

[2]

- Wind turbines and solar photovoltaic panels are interfaced to the power system through power converters which have very little over current capability.
- Fault current contribution from wind and solar photovoltaic generators is significantly less than an equivalent conventional synchronous generator due to the limited over current capability of wind and solar PV which reduces the short circuit level.

[1 mark each]

b) Consider a power system with a total number of buses N_b. A 3-phase fault occurs at bus k with a fault impedance Z_f. Derive the expression for the following in terms of the pre-fault voltage V_i^0 at bus i, appropriate elements of the bus impedance matrix $\boldsymbol{Z_{BUS}}$ and fault impedance Z_f. Assume that the row/column number of $\boldsymbol{Z_{BUS}}$ denotes the bus number and that Z_{mn} denotes the element in the mth row and nth column of $\boldsymbol{Z_{BUS}}$.

(i) Fault current I_f

[4]

- Change in bus voltages $\Delta \boldsymbol{V}$ due to the fault
  $\Delta \boldsymbol{V} = \boldsymbol{Z_{BUS}} \boldsymbol{J_f}$

  where $\boldsymbol{J_f} = [0 \quad 0 \quad \cdots \quad I_k = -I_f \quad \cdots \quad 0]^T$

- Voltage at the faulted (k^{th}) bus, V_k^f is given by:
  $V_k^f = V_k^0 - Z_{kk} I_f = Z_f I_f$

- Fault current I_f can be calculated from:
  $$I_f = \frac{V_k^0}{Z_{kk} + Z_f}$$

[1 mark each]

(ii) Voltage V_i^f at any bus i during the fault

[2]

- Voltage at any other bus i:
  $V_i^f = V_i^0 - Z_{ik} I_f$
  $$= V_i^0 - \frac{Z_{ik}}{Z_{kk} + Z_f} V_k^0, \; i = 1,2, \ldots N_b$$

  [1 mark each]

c) The bus impedance matrix ($\boldsymbol{Z_{BUS}}$) of a power system with 3 buses is given below where the row/column number of $\boldsymbol{Z_{BUS}}$ denotes the bus number.

Model Answers

$$\boldsymbol{Z_{BUS}} = \begin{bmatrix} j0.15 & j0.08 & j0.05 \\ j0.08 & j0.13 & j0.04 \\ j0.05 & j0.04 & j0.14 \end{bmatrix} \text{p.u.}$$

Consider a 3-phase fault at bus 2 with zero fault impedance, neglect the pre-fault currents and assume the pre-fault voltages at all the buses to be 1.0 p.u.

(i) Calculate the fault current magnitude (in p.u.).

[2]

Faulted bus no $k = 2$

Fault current

$$I_f = \frac{V_2^0}{Z_{22} + Z_f} = \frac{1.0}{j0.13} = -j7.69 \text{ p.u.}$$

$$|I_f| = 7.69 \text{ p.u.}$$

(ii) Calculate the voltage magnitude (in p.u) at each bus during the fault.

[3]

Voltages at each bus during fault are:

$$|V_1^f| = |V_1^0 - Z_{12} I_f| = 0.38 \text{ p.u.}$$

$$|V_2^f| = |V_2^0 - Z_{22} I_f| = 0 \text{ p.u.}$$

$$|V_3^f| = |V_3^0 - Z_{32} I_f| = 0.69 \text{ p.u.}$$

[1 mark each]

Some students used the incorrect element of the bus impedance matrix.

(iii) Find the Thevenin equivalent impedance of the power system at bus 1 and 3.

[1]

Thevenin equivalent impedance of the power system at a particular bus is given by the corresponding diagonal element of $\boldsymbol{Z_{BUS}}$

$$Z_{Th-1} = Z_{11} = j0.15 \text{ p.u.}$$

$$Z_{Th-3} = Z_{33} = j0.14 \text{ p.u.}$$

[0.5 mark each]

Model Answers

Question 5

a) The phase currents drawn by a star connected load with an ungrounded neutral point are $I_A = 1.05\angle 90°$ p.u. and $I_C = 0.99\angle 20°$ p.u. Calculate the magnitude (in p.u.) and phase angle (in degrees) of the positive- (A-B-C), negative- (A-C-B) and zero-sequence components of the load current.

[6]

$I_B = -(I_A + I_C) = 1.67\angle -123.82°$ p.u.

[1 mark]

$I_{A0} = I_{B0} = I_{C0} = \frac{1}{3}(I_A + I_B + I_C) = 0$

[1 mark]

$I_{A1} = \frac{1}{3}(I_A + \alpha I_B + \alpha^2 I_C) = 0.5\angle -1.39°$ p.u.

[1 mark]

$I_{B1} = \alpha^2 I_{A1} = 0.5\angle -121.39°$ p.u.

$I_{C1} = \alpha I_{A1} = 0.5\angle 118.61°$ p.u.

[1 mark]

$I_{A2} = \frac{1}{3}(I_A + \alpha^2 I_B + \alpha I_C) = 1.17\angle 115.15°$ p.u.

[1 mark]

$I_{B2} = \alpha I_{A2} = 1.17\angle -124.85°$ p.u.

$I_{C1} = \alpha^2 I_{A2} = 1.17\angle -4.85°$ p.u.

[1 mark]

Some students messed up the calculation despite using the correct approach. A few forgot to report the B and C phase currents of the sequence components.

b) For a line-to-ground (LG) fault in phase A, show that the magnitude of the fault current $|I_f|$ (in phase domain) is given by:

$$|I_f| = \left| \frac{3V_f}{Z_1 + Z_2 + Z_0 + 3Z_f} \right|$$

where Z_1, Z_2, Z_0 are the equivalent positive-, negative- and zero- sequence impedance, Z_f is the fault impedance and V_f is the pre-fault phase voltage.

[4]

Model Answers

Fault condition in phase domain
$I_B = I_C = 0, I_A = I_f, V_A = V_f = I_f Z_f$

[1 mark]

In sequence domain

$$\begin{bmatrix} I_{A0} \\ I_{A1} \\ I_{A2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}\begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}\begin{bmatrix} I_A \\ 0 \\ 0 \end{bmatrix}$$

$$I_{A0} = I_{A1} = I_{A2} = \frac{1}{3} I_A$$

[1 mark]

$$V_A = V_{A0} + V_{A1} + V_{A2} = I_f Z_f = 3 Z_f I_{A1}$$

[1 mark]

Hence, the sequence networks are connected in series across $3Z_f$

$$|I_f| = 3|I_{A1}| = \left| \frac{3V_f}{Z_1 + Z_2 + Z_0 + 3Z_f} \right|$$

[1 mark]

Some students just assumed the series connection of the sequence network instead of deriving it based on the relationship between the current and voltages in three sequence networks.

c) Two star-connected generators are connected in parallel to supply a static load through a line. The neutral point of one of the two generators is grounded through a reactor with $X_N = j0.1$ p.u. The neutral point of the other generator is left ungrounded. The values of the sequence reactance (in p.u.) of the generators and the line are given in Table 5.1.

Table 5.1: Sequence reactance of the generators and the line

	X_1	X_2	X_0
Generator	$j0.2$	$j0.18$	$j0.05$
Line	$j0.07$	$j0.07$	$j0.12$

A line-to-ground (LG) fault occurs on phase A at the bus where the static load is connected. The fault impedance is $Z_f = j0.03$ p.u. Assume the pre-fault voltages to be 1.0 p.u. and neglect the pre-fault current.

(i) Draw the three sequence networks for the LG fault condition showing the sequence reactance for individual components.

[2]

- Positive sequence network

Model Answers

1.0 p.u. voltage source in series with $j0.07$ (X_1 of line) and two $j0.2$ (X_1 of each generator) in parallel

- Negative sequence network

$j0.07$ (X_2 of line) and in series with two $j0.18$ (X_2 of each generator) in parallel

[1 mark]

- Zero sequence network

  $j0.12$ (X_0 of line), $j0.05$ (X_0 of the generator with grounded neutral) and $3 \times j0.1$ (X_N) in series

[1 mark]

Mistakes in zero-sequence network include not multiplying the neutral reactnace by 3, including/ignoring the zero-sequence reactance of both generators

(ii) Calculate the LG fault current (in p.u.).

[4]

$$Z_{1-eq} = \left(X_{g1} \parallel X_{g1}\right) + X_{l1} = j0.17 \text{ p.u.}$$

$$Z_{2-eq} = \left(X_{g2} \parallel X_{g2}\right) + X_{l2} = j0.16 \text{ p.u.}$$

$$Z_{0-eq} = X_{g0} + 3X_N + X_{l0} = j0.47 \text{ p.u.}$$

$$|I_f| = \left|\frac{3V_f}{Z_{1-eq} + Z_{2-eq} + Z_{0-eq} + 3Z_f}\right| = 3.37 \text{ p.u.}$$

[1 mark each]

Mistakes in zero-sequence network in the previous part affect d the results here. Some students didn't multiply the fault reactance by 3.

(iii) If the LG fault current was to be limited to half of that under the above situation, what value of neutral grounding reactance X_N (in p.u.) would be necessary?

[2]

$$\left|\frac{I_f'}{I_f}\right| = \left|\frac{Z_T}{Z_T + 3(X_N' - X_N)}\right| = 0.5$$

$$X_N' = \frac{Z_T}{3} + X_N = j0.397 \text{ p.u.}$$

[1 mark each]

(iv) Find the minimum value of the neutral grounding reactance X_N (in p.u.) to ensure that the fault current for an LG fault at the terminals of the generator is not more than the corresponding 3-phase fault current for zero fault impedance.

[2]

Model Answers

$$\left|\frac{3V_f}{(X_{g1} \parallel X_{g1}) + (X_{g2} \parallel X_{g2}) + X_{g0} + 3X_N}\right| \leq \left|\frac{V_f}{(X_{g1} \parallel X_{g1})}\right|$$

$$X_N \geq \frac{1}{3}\left[2(X_{g1} \parallel X_{g1}) - (X_{g2} \parallel X_{g2}) - X_{g0}\right] = j0.02 \text{ p.u.}$$

[1 mark each]

Some students considered the line reactance despite the fault being on the generator terminal.

Question 6

a) Define transient stability of power systems and state the two factors that primarily affect the transient stability.

[2]

- Transient stability is ability of the power system to maintain synchronism when subject to a large disturbance such as faults on transmission lines, transformers or buses and large change in generation/demand
- Transient stability depends primarily on the initial operating condition and the severity of the disturbance

[1 mark each]

b) A coal-fired power plant with round-rotor synchronous generators supplies a large power system through a transmission line. The voltages at the two ends of the line are regulated at 1.0 p.u. The reactance of the transmission line is 0.5 p.u. and its resistance and shunt charging are negligibly small. Neglect the resistance and reactance of the synchronous generators. Determine the maximum power (in p.u.) that can be transferred across the line in steady-state for the following conditions and sketch the corresponding power angle characteristics in each case.

(i) without any line compensation.

[2]

$$P_{max0} = \frac{V_1 V_2}{X} = 2.0 \text{ p.u.}$$

Sketch the following for $0 \leq \delta \leq \pi$
$P = 2 \sin \delta$

(ii) with 35% series compensation installed on the transmission line.

[2]

$$P_{max1} = \frac{V_1 V_2}{X(1-k)} = 3.08 \text{ p.u.}$$

Sketch the following for $0 \leq \delta \leq \pi$
$P = 3.08 \sin \delta$

Model Answers

Some students used incorrect relationship between the maximum power and compensation level.

(iii) with ideal phase angle compensation of -15° at one end without changing the effective voltage at that end.

[2]

$P_{max2} = \frac{V_1 V_2}{X} = 2.0$ p.u.

Sketch the following for $0 \leq \delta \leq \pi$
$P = 2\sin(\delta - 15°)$

(iv) with an ideal mid-point shunt reactive compensation installed exactly halfway through the length of the transmission line to maintain the mid-point voltage at 1.0 p.u.

[2]

$P_{max3} = \frac{V_1 V_m}{X/2} = 4.0$ p.u.

Sketch the following for $0 \leq \delta \leq \pi$
$P = 4\sin\frac{\delta}{2}$

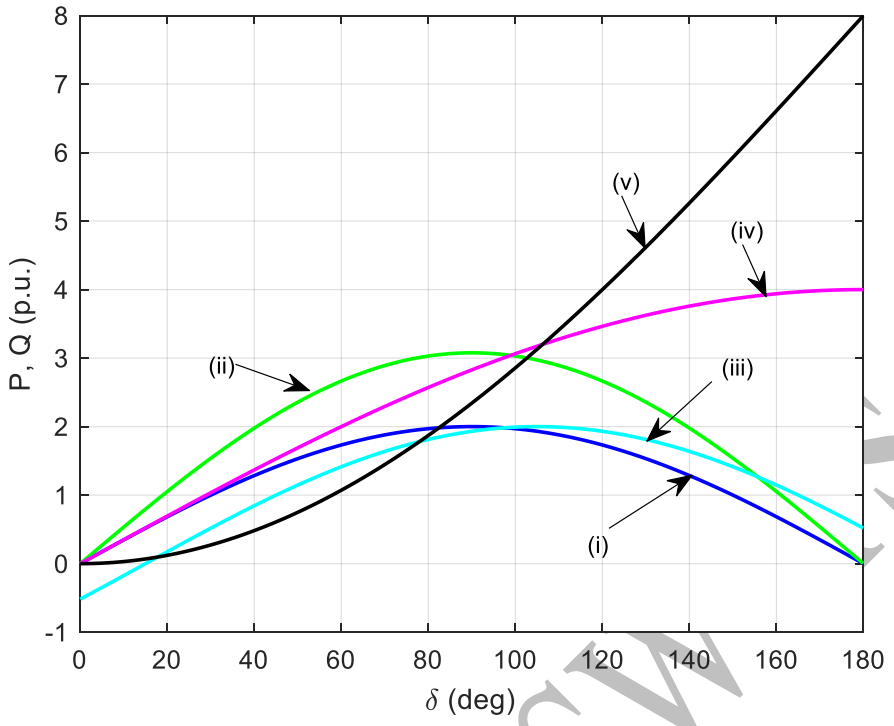
(v) for case (iv), calculate the reactive power (in p.u.) capacity required for the shunt compensator.

[2]

$Q_{cmax} = \frac{4V^2}{X} = 8.0$ p.u

Sketch the following for $0 \leq \delta \leq \pi$
$P = 8\left(1 - \cos\frac{\delta}{2}\right)$
Some students used incorrect relationship for maximum reactive power capacity.

Model Answers



[1 mark each for correct sketch and answer]

c) Consider the same context as in part (a) of this question. At a given point, the system is operating at 50% of its steady-state stability limit without any line compensation. Assuming the power output from the generators to be zero during a fault on the transmission line calculate the critical clearing angle (in degrees) for the following situations:

(i) without any line compensation. [4]

$$\delta_0 = \sin^{-1}\frac{P_m}{P_{max}} = 30°$$

Using the equal area criterion

$$\int_{\delta_0}^{\delta_{cr}} (P_m - 0)d\delta = \int_{\delta_{cr}}^{\pi-\delta_0} (P_{max}\sin\delta - P_m)d\delta$$

$$P_m(\delta_{cr} - \delta_0) = P_{max}[\cos\delta_{cr} - \cos(\pi - \delta_0)] - P_m(\pi - \delta_0 - \delta_{cr})$$

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 79.56°$$

[1 mark each]

Model Answers

(ii) with 35% series compensation but the generator output (i.e. power transfer through the line) remaining the same as in part (i).

[2]

$$\frac{P_m}{P_{max1}} = \frac{P_m(1-k)}{P_{max}} = 0.5 \times (1 - 0.35) = 0.325$$

$$\delta_1 = \sin^{-1}\frac{P_m}{P_{max1}} = 18.96°$$

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_1)\sin\delta_1 - \cos\delta_1] = 98.04°$$

[1 mark each]

In some cases, use of incorrect relationship between the maximum power and compensation level in 6b(ii) meant this part went wrong as well.

(iii) what is the power transfer (in p.u.) across the line with 35% series compensation if the critical clearing angle is the same as that in part (i).

[2]

As critical clearing angle (δ_{cr}) depends only on initial power angle (δ_0), $\frac{P_m{}'}{P_{max1}} = \sin\delta_0$ should be the same (i.e 0.5) as in part (i).

$$\frac{P_m{}'}{P_{max1}} = 0.5 \rightarrow P_m{}' = 0.5 \times P_{max1} = 1.54 \text{ p.u.}$$

[1 mark each]

Most students didn't realise the fact that the critical clearing angle depends only on the initial power angle which is a function of the ratio between the power transferred to the steady-state stability limit

Model Answers